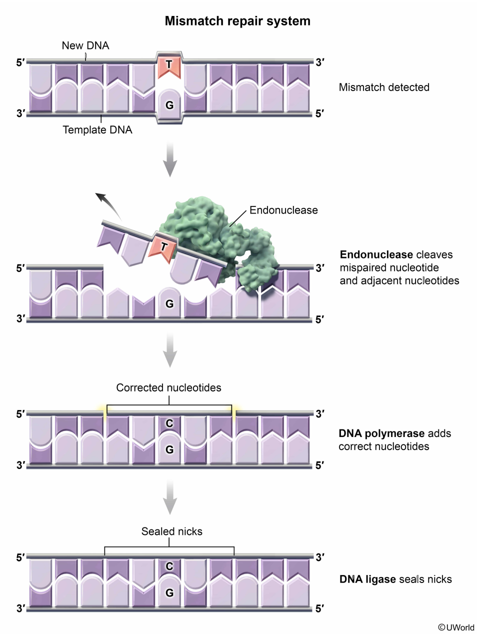
Which of the following nucleic acid complexes would undergo correction by the DNA mismatch repair system?

- A. 5'–UAGUCUUACAUUCC–3'
  Antisense 3'–ATCAGAATGTAAGG–5' [6%]
- B. 5'–GTGCCCACGATTCA–3'
  Antisense 3'–CACGGGTGCTAAGT–5' [1%]
- ✔ C. 5'–GCGCCACGATTTAA–3'
  Antisense 3'–CGCGGTGCTAAGTT–5' [64%]
- D. 5'–GGGCCCACGCUUUC–3'
  Antisense 3'–CCCGAGTGCGAAAG–5' [27%]

Correct
64% answered correctly

**Explanation:**




During DNA replication, DNA polymerase may **mistakenly incorporate** an **incorrect nucleotide** into the newly synthesized strand, creating a **base pair mismatch** between the daughter and the parent strands (ie, G-T and A-C mispairing instead of the proper G-C and A-T pairing). The presence of the mismatched base activates the **DNA mismatch repair (MMR) machinery**, in which a nuclease enzyme excises the erroneous nucleotide, as well as several nucleotides flanking either side of the incorrectly matched base, from the daughter strand. DNA polymerase then incorporates the appropriate nucleotides, and DNA ligase catalyzes the formation of new phosphodiester bonds to seal the gap between the strands.

In this scenario, the MMR machinery would operate on the DNA sequence with a pair of mismatched bases. The DNA complex in **Choice C** has a mismatch at the twelfth base pair that, once corrected, would produce an accurate complementary DNA strand.

**(Choice A)** The MMR machinery corrects only replication errors within DNA strands. This complex contains uracil (U) bases, which are found in RNA, *not* DNA. Accordingly, this complex would *not* activate the MMR machinery.

**(Choice B)** There is no mismatched base in these DNA strands; therefore, the MMR machinery would *not* be activated.

**(Choice D)** Although there is a mismatch in this choice at the fifth base pair, this complex contains U bases, which are found in RNA, *not* DNA. Therefore, this complex would *not* trigger the MMR machinery.

**Educational objective:**
Base pair mismatches occur when DNA polymerase incorporates an incorrect nucleotide into the newly synthesized strand. DNA mismatch repair is the process by which base pair mismatches between the daughter and the parent strands are corrected during DNA replication.

**Time spent:**0 sec
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